Answer **all** questions in the spaces provided.

1 **(a)** A soldier fires his semi-automatic rifle in a stationary position as shown in Fig. 1.1. Each bullet has a mass of 15 g and it leaves the gun with a velocity of 700 m s^{-1}.

Fig. 1.1

(i) If the rifle weighs 4.5 kg, calculate the velocity of the rifle when he fires a single shot.

velocity = ……. m s^{-1} [2]

(ii) If the soldier now fires 10 bullets horizontally in 2.0 s, calculate the average force exerted by the bullets on the rifle.

average force = ……. N [3]

For Examiner's Use

(b) Two balls X and Y having the same mass of 160 g collide head-on with each other as shown in Fig. 1.2.

Fig. 1.2

The graph in Fig. 1.3 shows the momentum of the balls before, during and after the collision.

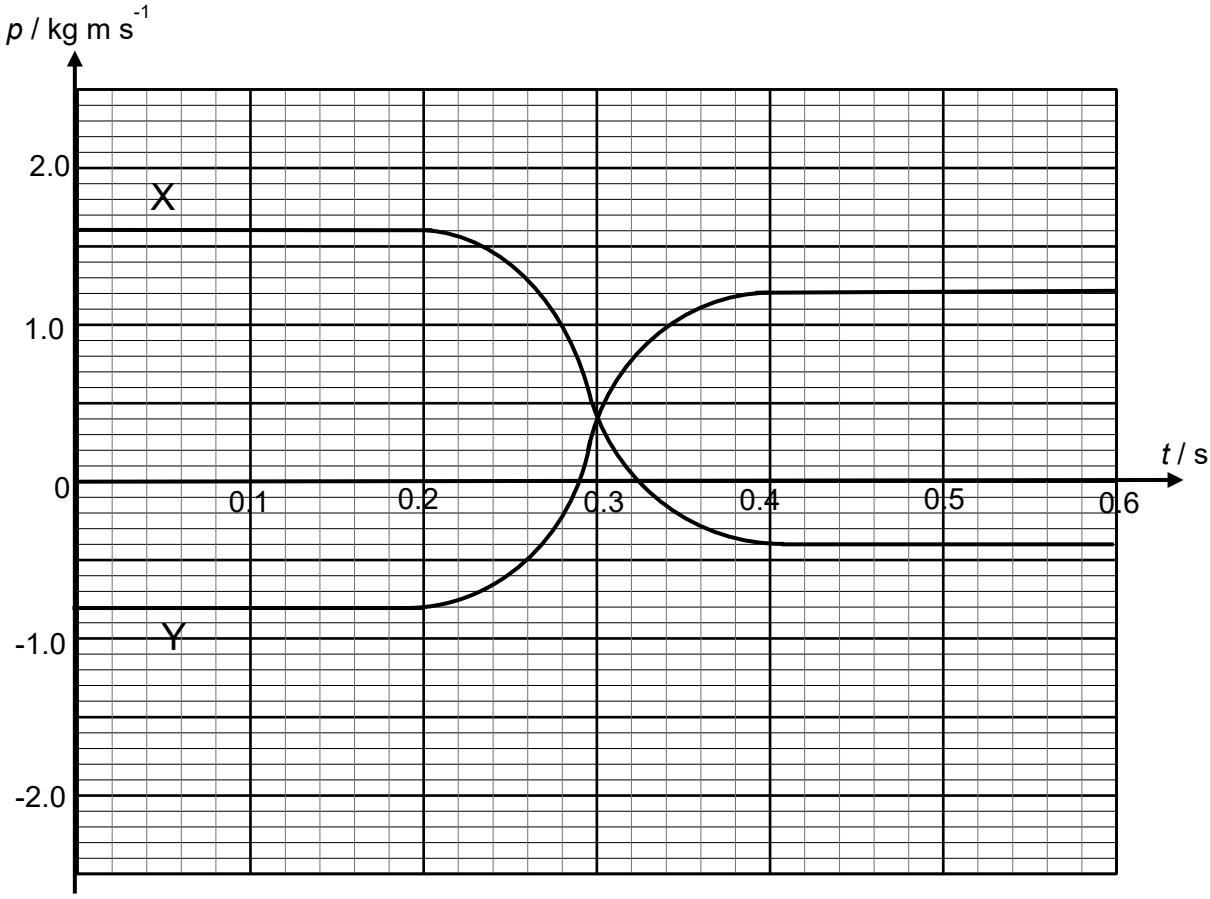

Fig. 1.3

(i) Show that the collision is inelastic.

[2]

For Examiner's Use

(ii) Calculate the impulse experienced by ball X due to the collision.

impulse = ……. N s [2]

(iii) Explain why both balls cannot lose all their kinetic energy at the same time during the collision.

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……..

……. [2]

For Examiner's Use

(c) Block B of mass 8.0 kg rests on a smooth 30° incline and is connected to block A of mass 12.0 kg via a frictionless pulley using an inextensible string as shown in Fig. 1.4. The blocks are released from rest and come to rest again momentarily when the spring is compressed by 1.2 m. The spring has a spring constant *k*.

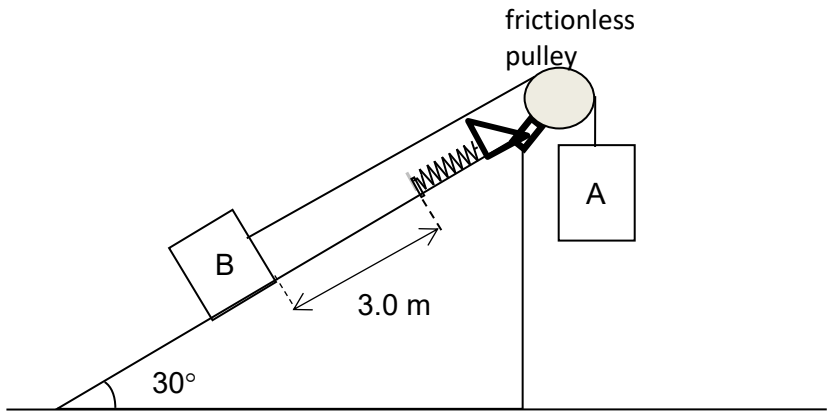

Fig. 1.4

(i) Describe the energy transformation from the time the blocks are released to the time when the blocks first come to rest.

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…… [3]

(ii) Calculate the value of *k.*

k = …… N m^{-1} [2]